An Improved Reference Device for Radiated Immunity Interlaboratory Comparison

Emrah Tas
Electomagnetic Compatibility Laboratory
Swiss Federal Institute of Metrology
Bern, Switzerland
emrah.tas@metas.ch

Dr. Frédéric Pythoud
Electomagnetic Compatibility Laboratory
Swiss Federal Institute of Metrology
Bern, Switzerland
frederic.pythoud@metas.ch

Beat Mühlemann
Electomagnetic Compatibility Laboratory
Swiss Federal Institute of Metrology
Bern, Switzerland
beat.muehlemann@metas.ch

***Abstract*— In order to organize interlaboratory comparisons and provide proficiency testing services in radiated immunity testing according to IEC 61000-4-3, a high quality reference device is developed. This device is adapted for the frequency range of 80 MHz to 6 GHz, and it is equipped with appropriate detecting mechanisms that record the required parameters for comparisons. This device is a significant improvement compared to a previous device already developed few years ago. The device architecture and the principles of operation are presented in detail, as well as a preliminary evaluation of its stability and linearity.**

Keywords— Equipment under test (EUT), Electromagnetic Compatibility (EMC), IEC 61000-4-3, IEC/ISO 13258, high frequency (HF), immunity, inter-laboratory comparison, proficiency testing, radiated, radio frequency (RF).

I. Introduction

Tests according to IEC 61000-4-3 intend to assess the immunity of an equipment under test (EUT) to radiated electromagnetic disturbance from different sources. This radiation might indeed disturb an EUT by coupling into the cables, into the housing of the EUT, or directly into the electronic prints of the EUT. This might generate disturbing voltages and currents that affect the correct functioning of the EUT. Nowadays, the radiated immunity tests are typically considered in frequency ranges from 80 MHz to 6 GHz to cover the current mobile and wireless communication standards.

The standard IEC 61000-4-3 precisely describes the setup, the calibration of the test place and the test procedure [3]. Similar to other electromagnetic compatibility (EMC) tests, the execution of this test requires not only appropriate infrastructure, but also technical skills. Even with the adequate experience and knowledge, the erroneous execution of the test is possible. The potential causes of the mistakes are numerous: incorrect cable layout, insufficient field homogeneity, inappropriate test levels, etc. In order to assess the testing capabilities of the laboratories and to demonstrate the testing quality according to IEC/ISO 17025, inter-laboratory comparisons or proficiency testing activities are required [4].

For the purpose of proficiency testing, various devices have already been designed for different EMC tests and the application of ISO 13528 for these EMC tests has been shown [5, 6, 7, 8, 9, 10]. For the radiated immunity testing, we previously introduced a reference device [1, 2]. This device suffered some drawbacks, such as the absence of the detection of the disturbance signal frequency and AM parameters, and the reduced frequency range up to 2.7 GHz. Moreover, in order to simulate a real EUT, a repeatable and predefined "failure" condition must be available in the reference device (See Section II for details).

In this publication, we present a significantly improved design of the reference device for radiated immunity interlaboratory comparisons. The device is based on the integration of commercially available components such high-speed USB-powered spectrum analyzers for frequency and signal level detection and a powerful mini PC for the system control and data recording.

This paper is organized as follows: Section II presents the device architecture, its mode of operation and characteristics. Testing scheme and the testing protocol as well as the analysis of the device stability and linearity are given in Section III. Finally, some concluding remarks and future work are presented in Section IV.

II. Device Architecture and Characteristics

This improved reference device for interlaboratory comparisons in radiated immunity is based on a mini PC and three high-speed real-time USB Spectrum Analyzers covering the frequency range from 80 MHz to 6 GHz. The picture of the device is given in Figure 1. Similar to the previous device [1], we again implemented three different coupling channels to represent the coupling mechanisms on different interfaces (See Figure 2).

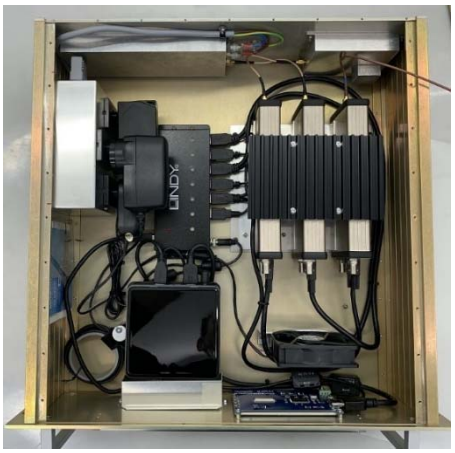


Figure 1. Overview of the Reference Device

Channel 1 of the device records the symmetrical signal between L and N terminals of the mains via a coupling system. The transmission characteristics (S21) of the implemented coupling system vs. frequency up to 1 GHz are given in Figure

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3. The coupling system has been optimized for the frequencies up to 300 MHz.

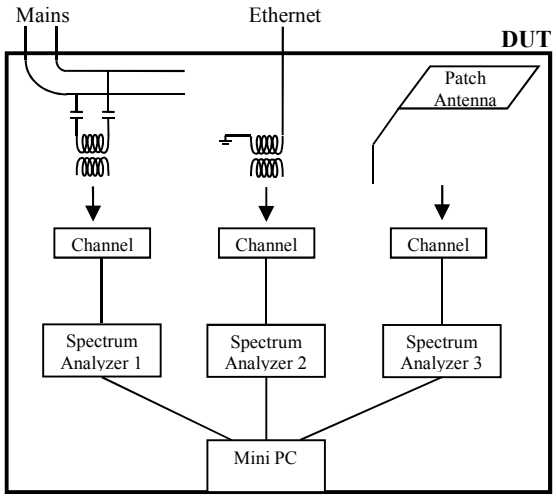


Figure 2. Coupling mechanisms of the Reference Device

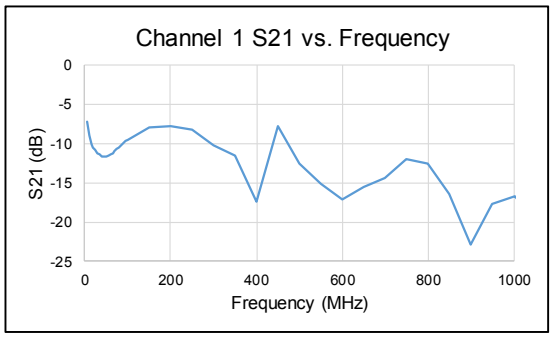


Figure 3. Transmission Characteristics (S21) of Coupling System on Mains (Channel 1) vs. Frequency

On Channel 2, the common mode signal from the unshielded Ethernet interface is recorded. According to the measurements of the transmission of the coupling mechanism for this channel (see Figure 4), the coupling works correctly up to 3 GHz. Based on further stability analyses, we reduced the measurements on this channel to the frequency range from 80 MHz to 1.5 GHz.

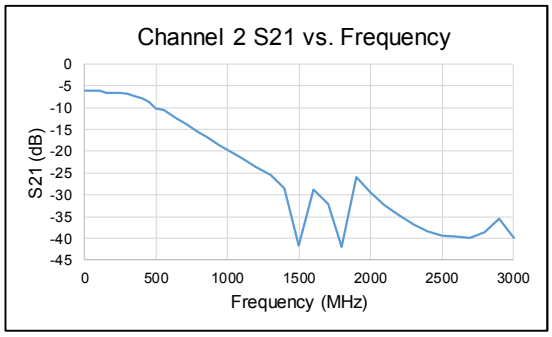


Figure 4. Transmission Characteristics (S21) of Coupling System on Ethernet (Channel 2) vs. Frequency

A commercial patch antenna (Taoglas Maximus) is placed under the lid of the device and it is used for the measurements on Channel 3. This antenna is specified by the manufacturer for the frequency range from 700 MHz to 6 GHz, with 5 dBi peak gain and -2 dB average gain [11]. Tests with the antenna showed stable measurements starting from 500 MHz. The radiated field with relatively short wavelengths are measured over this interface. The measurements on Channel 3 are thus optimized for the frequency range from 500 MHz to 6 GHz.

The spectrum analyzers are Signal Hound BB60C, which is a high-speed real-time USB Spectrum Analyzer. They cover the frequency range from 9 kHz to 6 GHz, in agreement with the frequency range of IEC 61000-4-3 [12]. They have SMA interface for RF signal input and a USB interface for real-time data transmission and for 5V power supply. The device comes with its own proprietary software to display the measured signal and perform additional operations. Moreover, a software development kit is also available for further custom software designs.

In order to evaluate the accuracy of the spectrum analyzers, their frequency response for signal levels from -20 dBm to -60 dBm with 10 dB steps has been measured (see Figure 5). This figure shows that the accuracy and linearity of the spectrum analyzer are good, and the absolute accuracy is practically better than 1 dB.

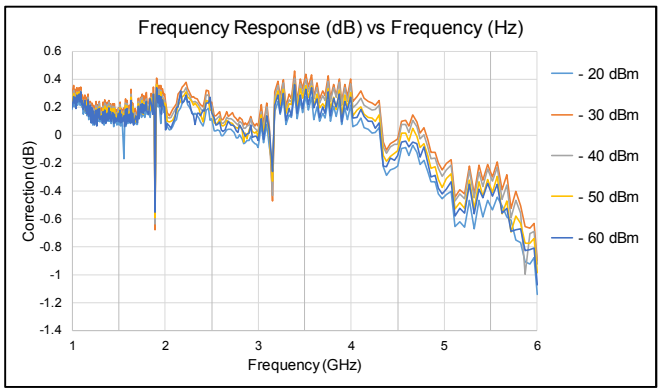


Figure 5. Required Correction on Analyzer Reading vs. Frequency for Different Input Signal Levels

The reference device is programmed in fact as a data logger. The custom software on the mini PC controls each spectrum analyzer by setting its tuned frequency, trigger threshold, reference signal level and other relevant parameters. Whenever a spectrum analyzer detects a significant signal on the corresponding channel, the software captures the signal level and frequency and stores it on a removable USB hard drive. Moreover, by using a low resolution bandwidth (RBW), the device is also capable of demodulating the AM signal, gathering its AM index and frequency. However, this mode slows down the acquisition process, and therefore, in our implementation, AM detection is limited only to 9 selected frequencies: 110 MHz, 500 MHz, 800 MHz, 1.2 GHz, 1.9 GHz, 2.7 GHz, 3.2 GHz, 4.2 GHz and 5.4 GHz. For these frequency points, the RBWs of the analyzers are temporarily reduced to 300 Hz, whereas for other frequencies, RBWs are set to 10 kHz.

For the control of the device, a touch screen has been placed on the front panel. In addition to its limited functions (i.e. start/stop of a measurement and switching the device off), this front panel also simulates a "failure". The functioning of the EUTs may be impaired during immunity tests. In order to realize such a case, the background color of the touchscreen turns from gray into red when a predefined test condition occurs (For example when the signal level measured on

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Channel 1 is in between -15 dBm and -5 dBm in the frequency range from 200 MHz to 300 MHz). The test operator should note the frequencies where this "failure" occurs. This is also an indication of a successful testing in measurement comparisons.



(a)



(b)

Figure 6. Front Panel of the Device
(a) Metallic Cover Closed (b) Metallic Cover Open

The device has a metallic housing in order to protect its sensible electronic components from the radiated field. The touch screen and the removable USB hard drive are protected by a metallic window with screws and a transparent foil having metallic wires on it so that the device can be observed for the above-mentioned "failure" situation (Figure 6). Moreover, the opening for the patch antenna is covered with a protective polytetrafluoroethylene (PTFE) surface in order to permit the reception of the incoming radiation by the antenna with minimum attenuation. Furthermore, the shield of the points, where the SMA cables of the coupling paths are connected to the corresponding spectrum analyzers, are grounded to the device housing. With these solutions, the effective shielding and the Faraday cage functionality of the device are ensured for the whole frequency range. Although the device is designed for the industrial test environments of 10 V/m, it was tested up to 20 V/m to ensure its robustness.

The device also includes an auxiliary equipment (AE) with Ethernet interface to provide a defined coupling to Channel 2 (Figure 7). This AE is a “Dummy” that reproduces the ground connections of a typical AE. It has five LED indicators for the successful connection with the reference device. The test setup has to be prepared with this AE according to IEC 61000-4-3.

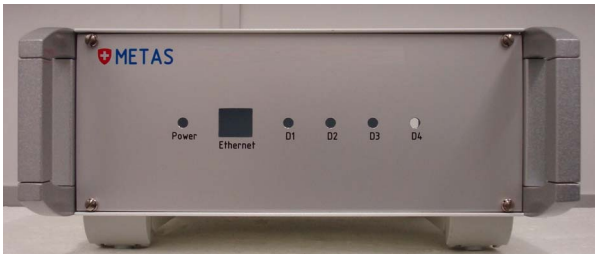


Figure 7. The auxiliary equipment for the radiated immunity reference device

III. Device Specification and Preliminary Evaluation

A. Device and Testing Specifications

The reference device is capable of recording the strength and the frequency of the electromagnetic field in whole frequency range 80 MHz – 6 GHz on all the integrated channels. Since all dimensions of the device is less than 1.0 m, full illumination method according to IEC 61000-4-3 is possible, even at frequencies above 1 GHz.

A typical test protocol as shown in Table 1 is derived from IEC 61000-4-3.

Test Specification	Units	Polarization	Turntable Position
80 to 6000 10 80	MHz V/m % AM (1 kHz)	Horizontal, Vertical	0□, 90□, 180□, 270□

Table 1. Testing Protocol

B. Device Stability

In order to assess the stability of the device, measurements according to protocol of Table 1 have been performed. Since the complete analysis of stability includes a total of 24 combinations of the field polarization, turntable position and the evaluated channel, only a partial evaluation is depicted here for a given position and polarization.

Figure 8 depicts the test setup for Position 90° and horizontal polarization. The measured signal levels vs. frequency on three channels for three different measurements sets and the corresponding standard deviations are given in Figures 9-14.

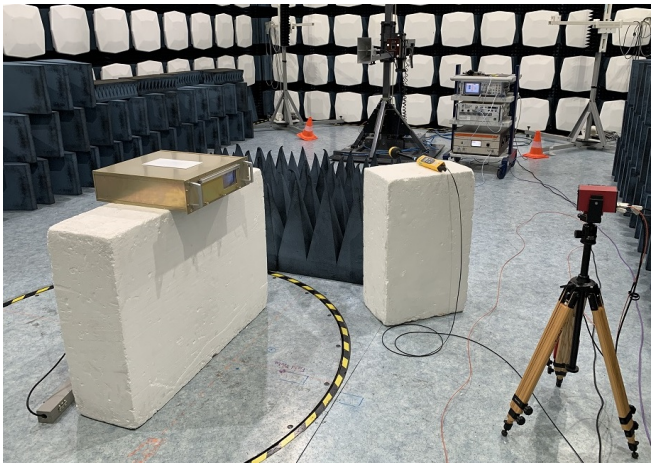


Figure 8. Test Setup for Position 90° and Horizontal Polarization

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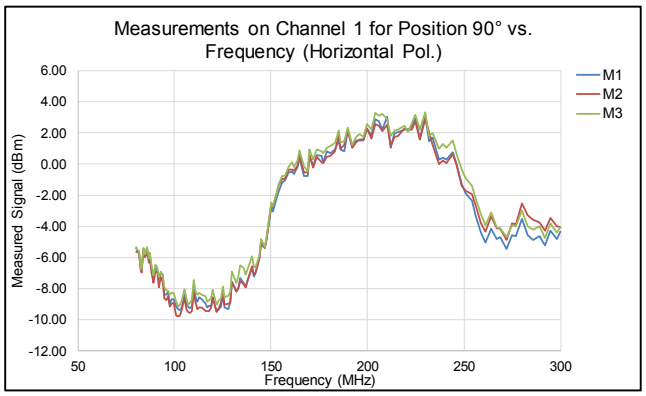


Figure 9. Signal Level on Channel 1 for Position 90° vs. Frequency (Horizontal Polarization) ("M": Measurement)

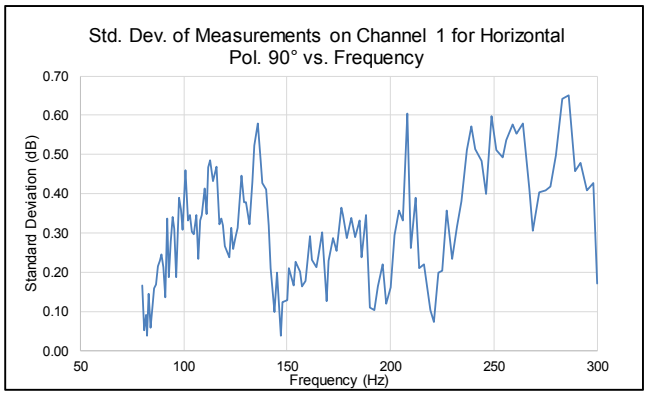


Figure 10. Standard Deviation of the Measurements on Channel 1 for Position 90° vs. Frequency (Horizontal Polarization)

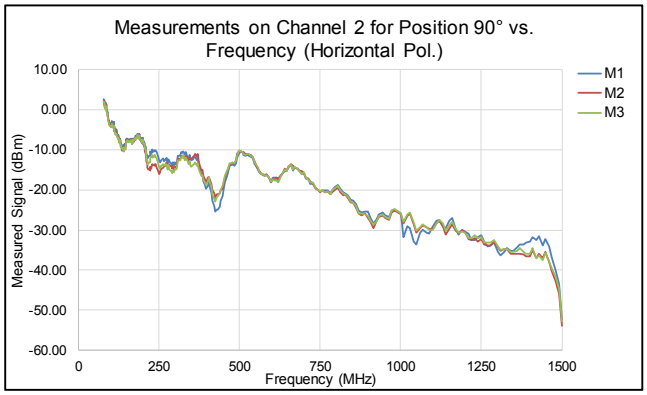


Figure 11. Signal Level on Channel 2 for Position 90° vs. Frequency (Horizontal Polarization) ("M": Measurement)

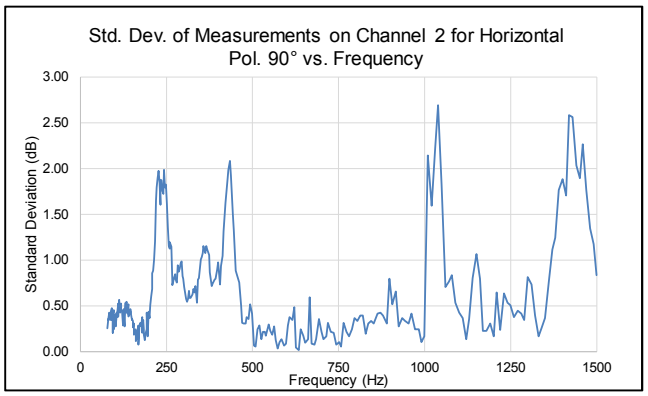


Figure 12. Standard Deviation of the Measurements on Channel 2 for Position 90° vs. Frequency (Horizontal Polarization)

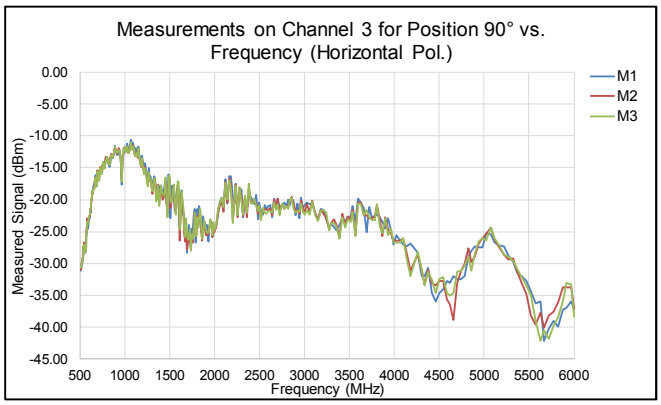


Figure 13. Signal Level on Channel 3 for Position 90° vs. Frequency (Horizontal Polarization) ("M": Measurement)

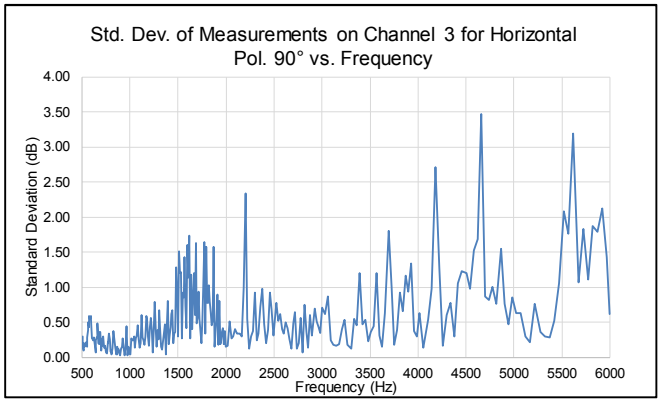


Figure 14. Standard Deviation of the Measurements on Channel 3 for Position 90° vs. Frequency (Horizontal Polarization)

When this analysis is repeated for every combination of the position and the field polarization, the observed maximum standard deviations are given in Table 2.

Position and Polarization	Maximum Std. Dev. on Channel 1 (dB)	Maximum Std. Dev. on Channel 2 (dB)	Maximum Std. Dev. on Channel 3 (dB)
0□ - Hor.	2.02	4.07	1.80
0□ - Ver.	1.20	1.97	3.34
90□ - Hor.	0.65	2.69	3.47
90□ - Ver.	1.67	3.91	3.70
180□ - Hor.	1.53	3.38	4.97
180□ - Ver.	2.46	2.24	2.34
270□ - Hor.	1.86	3.46	3.12
270□ - Ver.	2.07	4.10	3.90

Table 2. Maximum Standard Deviation of Measurements for All Cases

C. Discussion on Device Stability

The calibration of the test area has the intrinsic measurement uncertainty of 6 dB [3]. This uncertainty already creates possible deviations between the measurement positions.

On the other hand, one of the most important requirement of a suitable reference device is to deliver repeatable measurements under the same test conditions. In order to fulfill this requirement, the resonances occurring in the

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measurements should be avoided. Since the device has metallic housing and the antenna is placed on the lid surrounded by a metallic surface, some resonances are inevitable. This problem can be avoided by removing the frequencies of the significant resonances from the measurement comparison. The device delivers stable measurements in a very wide portion of the whole frequency range, which can safely be used for measurement comparison purposes.

Moreover, it was observed that the AM frequency and index detection could be performed with a precision of about 1% at the given frequencies. Furthermore, during these tests, it was observed that none of the device components suffered any functional impairment, which proves the shielding effectiveness of the device housing.

Considering these observations, it can be concluded that the device delivers stable and repeatable measurement results, which makes it suitable to be used as a reference device in radiated immunity measurement comparisons according to IEC 61000-4-3.

D. Device Linearity

In order to evaluate the linearity of the device, the measurements for the Position 90° and horizontal polarization for the field strengths of 3 V/m and 10 V/m in the frequency range from 80 MHz to 1 GHz have been compared. The difference between these field strengths is expected to yield 10.46 dB in the power readings on all channels. Figures 15-20 show the measurements on each channel and the corresponding difference.

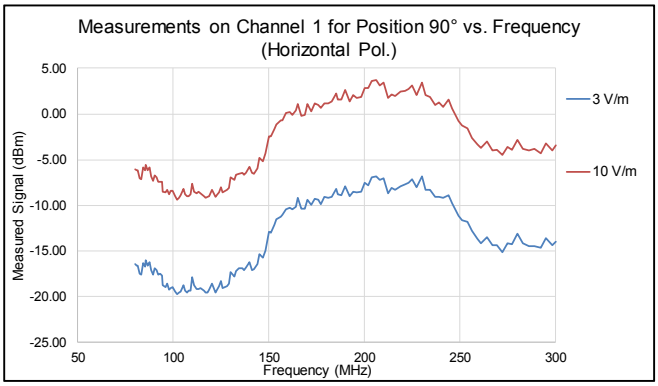


Figure 15. Signal Level on Channel 1 for Position 90° vs. Frequency (Horizontal Polarization) for 3 V/m and 10 V/m cases

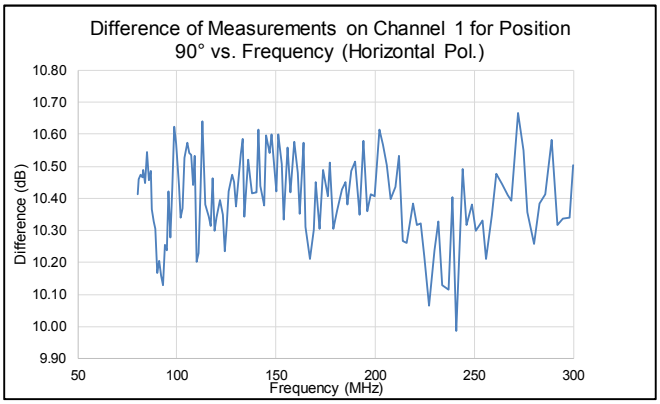


Figure 16. Difference of the Measurements on Channel 1 for Position 90° vs. Frequency (Horizontal Polarization) for 3 V/m and 10 V/m cases

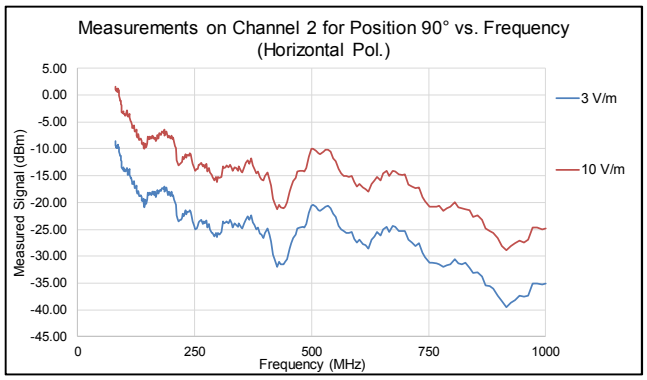


Figure 17. Signal Level on Channel 2 for Position 90° vs. Frequency (Horizontal Polarization) for 3 V/m and 10 V/m cases

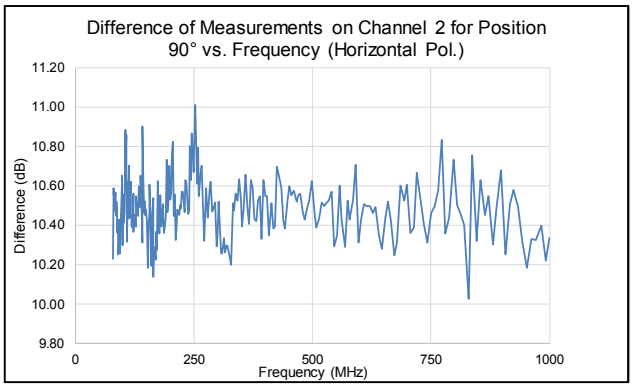


Figure 18. Difference of the Measurements on Channel 2 for Position 90° vs. Frequency (Horizontal Polarization) for 3 V/m and 10 V/m cases

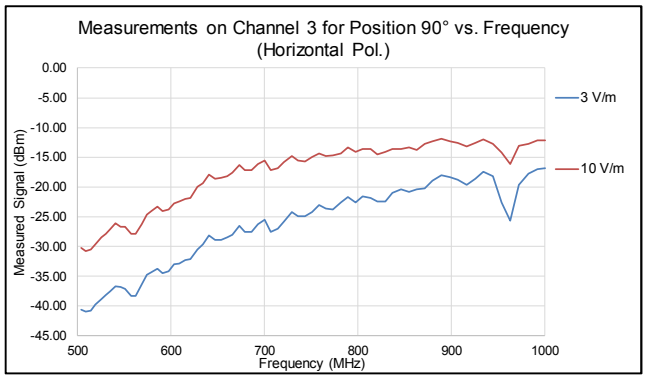


Figure 19. Signal Level on Channel 1 for Position 90° vs. Frequency (Horizontal Polarization) for 3 V/m and 10 V/m cases

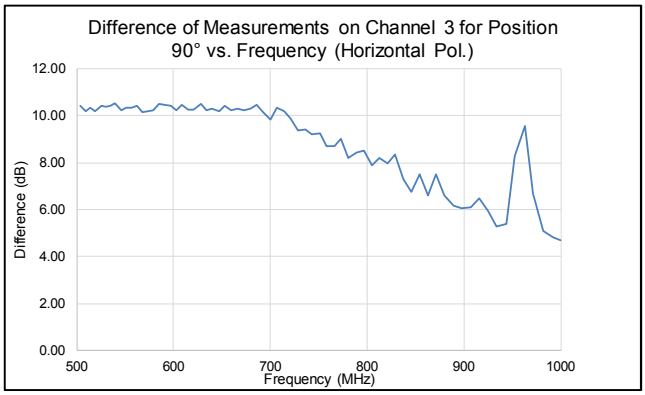


Figure 20. Difference of the Measurements on Channel 3 for Position 90° vs. Frequency (Horizontal Polarization) for 3 V/m and 10 V/m cases

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E. Discussion on Device Linearity

The above figures show that the linearity (i.e. 10.46 dB difference on the readings) is achieved with a good precision (± 0.6 dB) on channels 1 and 2, but on channel 3, the linearity is severely distorted after 700 MHz. This is due to the reference level setting of the spectrum analyzer on Channel 3. Since the reference level was set to -15 dBm, when the measured signal level surpasses this value, which happens after 700 MHz, the spectrum analyzer goes into compression and saturation. According to the spectrum analyzer manufacturer, the reference level should be set 5 dB higher than the highest expected power level. Therefore, this problem can be solved by increasing the reference level to around -8 dBm for the measurements in the whole frequency range from 80 MHz to 6 GHz.

IV. Conclusion and Future Work

This work constitutes not only a concept check for a better design of a reference device for radiated immunity interlaboratory comparisons according to IEC 61000-4-3, but also the verification of the realized device with measurements under real test conditions. The first results given here show that the stability and the linearity of the reference device are very good. However, the intrinsic 6 dB field uniformity tolerance defined by the standard would yield higher deviations in the measurement comparisons. The standard deviation of the proficiency testing will be calculated by considering all these relevant parameters.

The few imperfections found out during this research are relatively easy to overcome. This new device has numerous advantages: a robust construction, autonomous functioning (i.e. no need to perform reading with external instruments), capability for automatically detecting the frequency of the disturbance signal, simulating a predefined "failure", which in turn can be used as a performance metric in the measurement comparisons, etc.

As some software problems have been observed during these measurements, a more robust software implementation is planned for the future. Moreover, the linearity aspects will be investigated more deeply in the whole frequency range. After these improvements, a Swiss national measurement comparison will be conducted, which would bring more experience with the device.

Acknowledgment

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